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CSE C2 37

Lab 7: Form Processing using Django – Part I

Lab Assignments

urls.py (common):

"""

URL configuration for myproject project.

The `urlpatterns` list routes URLs to views. For more information please see:

<https://docs.djangoproject.com/en/5.2/topics/http/urls/>

Examples:

Function views

1. Add an import: from my_app import views
2. Add a URL to urlpatterns: path("", views.home, name='home')

Class-based views

1. Add an import: from other_app.views import Home
2. Add a URL to urlpatterns: path("", Home.as_view(), name='home')

Including another URLconf

1. Import the include() function: from django.urls import include, path
2. Add a URL to urlpatterns: path('blog/', include('blog.urls'))

"""

from django.contrib import admin

from django.urls import path, include

```
urlpatterns = [  
    path('studentform/', include('studentform.urls')),  
    path('carform/', include('carform.urls')),  
    path('formex/', include('formex.urls')),  
    path('admin/', admin.site.urls),  
    path('webapp/', include('webapp.urls')),  
    path('formapp/', include('formapp.urls')),  
    path('ex1app/', include('ex1app.urls')),  
    path('ex2app/', include('ex2app.urls')),  
    path('homeex1app/', include('homeex1app.urls')),  
    path('basic_calc/', include('basic_calc.urls')),  
    path('magazine/', include('magazine.urls')),  
    path('book/', include('book.urls')),  
    path('label/', include('label.urls')),  
]
```

1) Develop a web application using Django framework to demonstrate the transfer of multiple parameters between web pages. User should be presented with a dropdown list containing car manufacturers, a text box which takes model name of the manufacturer and a submit button. On submitting the web page, the user is forwarded to a new page. This new page should display the selected car manufacturer name and the model name.

forms.py:

```
from django import forms

class CarForm(forms.Form):
    CAR_CHOICES = [
        ('Toyota', 'Toyota'),
        ('Honda', 'Honda'),
        ('Ford', 'Ford'),
        ('BMW', 'BMW'),
        ('Hyundai', 'Hyundai'),
    ]

    manufacturer = forms.ChoiceField(
        choices=CAR_CHOICES,
        label="Select Car Manufacturer"
    )

    model_name = forms.CharField(
        max_length=100,
        label="Enter Model Name"
    )
```

views.py:

```
from django.shortcuts import render
from .forms import CarForm

def home_view(request):
    if request.method == "POST":
        form = CarForm(request.POST)
        if form.is_valid():
            manufacturer = form.cleaned_data['manufacturer']
            model_name = form.cleaned_data['model_name']

            return render(request, "result.html", {
                'manufacturer': manufacturer,
```

```
        'model_name': model_name
    })
else:
    form = CarForm()

return render(request, "home.html", {'form': form})
```

urls.py:

```
from django.urls import path
from . import views

urlpatterns = [
    path("", views.home_view, name='home_view')
]
```

home.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>Car Selection</title>
</head>
<body>
    <h2>Select Car Details</h2>

    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <input type="submit" value="Submit">
    </form>

</body>
```

```
</html>
```

result.html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Selected Car</title>
```

```
</head>
```

```
<body>
```

```
    <h2>Car Details Submitted</h2>
```

```
    <p><strong>Manufacturer:</strong> {{ manufacturer }}</p>
```

```
    <p><strong>Model Name:</strong> {{ model_name }}</p>
```

```
</body>
```

```
</html>
```

output:



The screenshot shows a web browser window with the title "Car Selection". The address bar displays the URL "http://127.0.0.1:8000/carform/". The page content is titled "Select Car Details". It features a form with two input fields: "Select Car Manufacturer:" with a dropdown menu showing "Toyota", and "Enter Model Name:" with a text input field containing "Corolla". A "Submit" button is located at the bottom left of the form.



2) Create a page firstPage.html with two TextBoxes [Name, Roll], DropDownList [Subjects], and a button. Create another page secondPage.html with a label and a button. When the user clicks the button in first Page, he should be sent to the second page and display the contents passed from first page in the label. The button in second page should navigate the user back to firstPage. Use Django sessions to transfer information.

forms.py:

```
from django import forms
```

```
class StudentForm(forms.Form):
```

```
    SUBJECT_CHOICES = [  
        ('Maths', 'Maths'),  
        ('Physics', 'Physics'),  
        ('Chemistry', 'Chemistry'),  
        ('Computer Science', 'Computer Science'),  
    ]
```

```
name = forms.CharField(max_length=100, label="Name")
roll = forms.CharField(max_length=20, label="Roll")
subject = forms.ChoiceField(choices=SUBJECT_CHOICES, label="Subject")
```

views.py:

```
from django.shortcuts import render, redirect
```

```
# Create your views here.
```

```
from .forms import StudentForm
```

```
def first_page(request):
```

```
    if request.method == "POST":
```

```
        form = StudentForm(request.POST)
```

```
        if form.is_valid():
```

```
            # Store data in session
```

```
            request.session['name'] = form.cleaned_data['name']
```

```
            request.session['roll'] = form.cleaned_data['roll']
```

```
            request.session['subject'] = form.cleaned_data['subject']
```

```
            return redirect('second')
```

```
    else:
```

```
        form = StudentForm()
```

```
    return render(request, "firstPage.html", {'form': form})
```

```
def second_page(request):
```

```
    name = request.session.get('name')
```

```
    roll = request.session.get('roll')
```

```
subject = request.session.get('subject')

context = {
    'name': name,
    'roll': roll,
    'subject': subject
}

return render(request, "secondPage.html", context)
```

urls.py:

```
from django.urls import path
from . import views

urlpatterns = [
    path("", views.first_page, name='first'),
    path('second/', views.second_page, name='second'),
]
```

firstPage.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>First Page</title>
</head>
<body>

<h2>Enter Student Details</h2>

<form method="POST">
    {% csrf_token %}
```

```
    {{ form.as_p }}
    <input type="submit" value="Submit">
</form>

</body>
</html>
```

secondPage.html:

```
<!DOCTYPE html>

<html>

<head>
    <title>Second Page</title>
</head>
<body>

<h2>Details Submitted</h2>

<p>
    Name: {{ name }} <br>
    Roll: {{ roll }} <br>
    Subject: {{ subject }}
</p>

<form action="{% url 'first' %}">
    <input type="submit" value="Go Back">
</form>

</body>
</html>
```

output:

First Page

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http://127.0.0.1:8000/studentform/

Enter Student Details

Name:

Roll:

Subject:

Second Page

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http://127.0.0.1:8000/studentform/second/

Details Submitted

Name: Aditya Mathpal

Roll: 37

Subject: Computer Science